

Vemuri's Exceptional Times: A One-Dimensional Literature Completion

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Status. literature_already_answered (one-dimensional exceptional-time completion).

Source Question

For $a > 0$, let $E_a^\infty(1)$ be the class of functions on $\mathbb{R}$ for which both the function and its Fourier transform have Gaussian decay of rate a . Let $\Phi_f(\cdot, t)$ denote the harmonic oscillator evolution with initial datum f .

Kulikov, Oliveira, and Ramos state Vemuri's conjecture on PDF page 3 of arXiv:2202.11193: if

$$f \in E_{\tanh(2\alpha)}^\infty(1),$$

then

$$\Phi_f(\cdot, t) \in E_{\tanh(\alpha)}^\infty(1) \quad \text{for every } t \in \mathbb{R}.$$

Their Theorem 1.1 on PDF page 4 proves the corresponding statement in dimension n except when

$$t \in \left\{ \frac{1}{16} + \frac{k}{8} : k \in \mathbb{Z} \right\}.$$

Thus, in particular, the exceptional times remained open for the original one-dimensional conjecture [1].

Supporting Answer

Radchenko and Ramos explicitly identify this gap. On PDF page 2 of arXiv:2305.18546 they write that Kulikov–Oliveira–Ramos proved Vemuri's conjecture except at those times and that the purpose of their note is to prove the remaining cases. Their Theorem 1 states that, in dimension one,

$$f \in E_{\tanh(2\alpha)}^\infty \implies |\Phi_f(x, t)| \leq C e^{-\pi \tanh(\alpha) |x|^2} \quad (x, t \in \mathbb{R}).$$

Applying the same all-time estimate after the quarter-period Fourier shift gives the corresponding bound for the Fourier transform of $\Phi_f(\cdot, t)$. Hence the full $E_{\tanh(\alpha)}^\infty$ conclusion holds for every time, including the exceptional progression [2].

This is an explicit literature completion: the supporting authors cite the source paper and state that they are closing exactly its remaining cases.

Scope Limitations

The supporting theorem is one-dimensional. This packet does not claim a completion of the source paper’s higher-dimensional exceptional-time case. It also does not answer the stronger time-dependent Gaussian exponent conjecture formulated later in the source (Conjecture 3.1 in the published numbering), nor the general endpoint conversion question for Gaussian L^2 bounds.

Search Evidence

The run indexes were searched for arXiv:2202.11193, arXiv:2305.18546, Vemuri’s conjecture, and the exceptional-time phrase; no prior packet was found. A bounded web/arXiv search on 2026-07-21 located arXiv:2305.18546 and verified the explicit source-to-answer citation and theorem match.

References

- [1] A. Kulikov, L. Oliveira, and J. P. G. Ramos, *On Gaussian decay rates of harmonic oscillators and equivalences of related Fourier uncertainty principles*, arXiv:2202.11193; Rev. Mat. Iberoam. **40** (2024), 481–502.
- [2] D. Radchenko and J. P. G. Ramos, *Sharp Gaussian decay for the one-dimensional harmonic oscillator*, arXiv:2305.18546; Proc. Amer. Math. Soc. **153** (2025), 1985–1991.